Lab Report - 2

Group - 04

Section - 27 - A

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Lab Experiment - 4

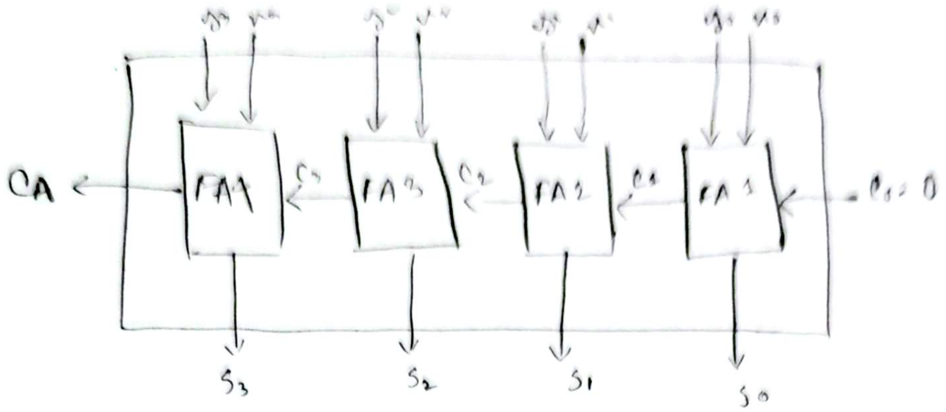
Experiment Name: Design and Implementation of 4-bit Parallel Binary adder.

Required components:

1) IC 7408
2) IC 7432
3) IC 7486
4) IC 7483

Experimental Setup: ....

Experiment - 1



4- bit parallel Adder

	A	B	Cin	C4	S4	S3	S2	S1
a	0000	1000	0	0	1	0	0	0
b	1100	1101	0	1	1	0	0	1
c	1110	0011	0	1	0	0	0	1
d	1111	1111	0	1	1	1	1	0

4 - bit Parallel Adder cum Subtractor

	A	B	Cin	C4	S4	S4	S2	S1
a	1100	1000	1	0	0	1	0	0
b	1100	1101	0	1	1	0	0	1
c	1110	0011	1	0	1	0	1	1
d	1111	1111	0	1	1	1	1	0

Experiment 5: Implementation of 4 bit Magnitude comparator

Required components

1) IC 7408

2) IC 7432

3) IC 7404

4) IC 4077

Experimental Setup....

Experiment 06 : Design circuits using Encoder & Decoder

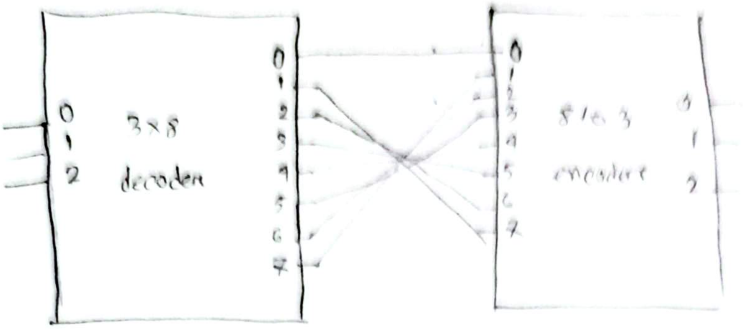
Required Components

1) IC 74138

2) IC 74148

Experimental Setup : . . .

Experiment-6
Group-04



Inputs				Expected Outputs				Active low Outputs		
Minterm	C	B	A	Minterm	D_2	D_1	D_0	D_2	D_1	D_0
0	0	0	0	0	0	0	0	1	1	1
1	0	0	1	7	1	1	1	0	0	0
2	0	1	0	6	1	1	0	0	0	1
3	0	1	1	5	1	0	1	0	1	0
4	1	0	0	4	1	0	0	0	1	1
5	1	0	1	3	0	1	1	1	0	0
6	1	1	0	2	0	1	0	1	0	1
7	1	1	1	1	0	0	1	1	1	0

Experiment 07 : Function Implementation using MUX

Required components :

1) IC 74153
2) IC 7408
3) IC 7432
4) IC 7409

Experimental setup....

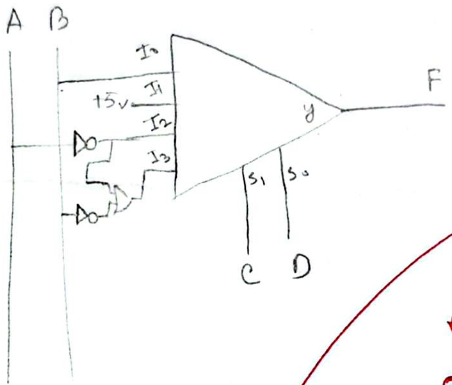
	I0	I1	I2	I3
A'B'	0	1	2	3
A'B	4	5	6	7
AB'	8	9	10	11
AB	12	13	14	15

$I0 = B$

$I1 = 1$

$I2 = A'$

$I3 = A' + B'$

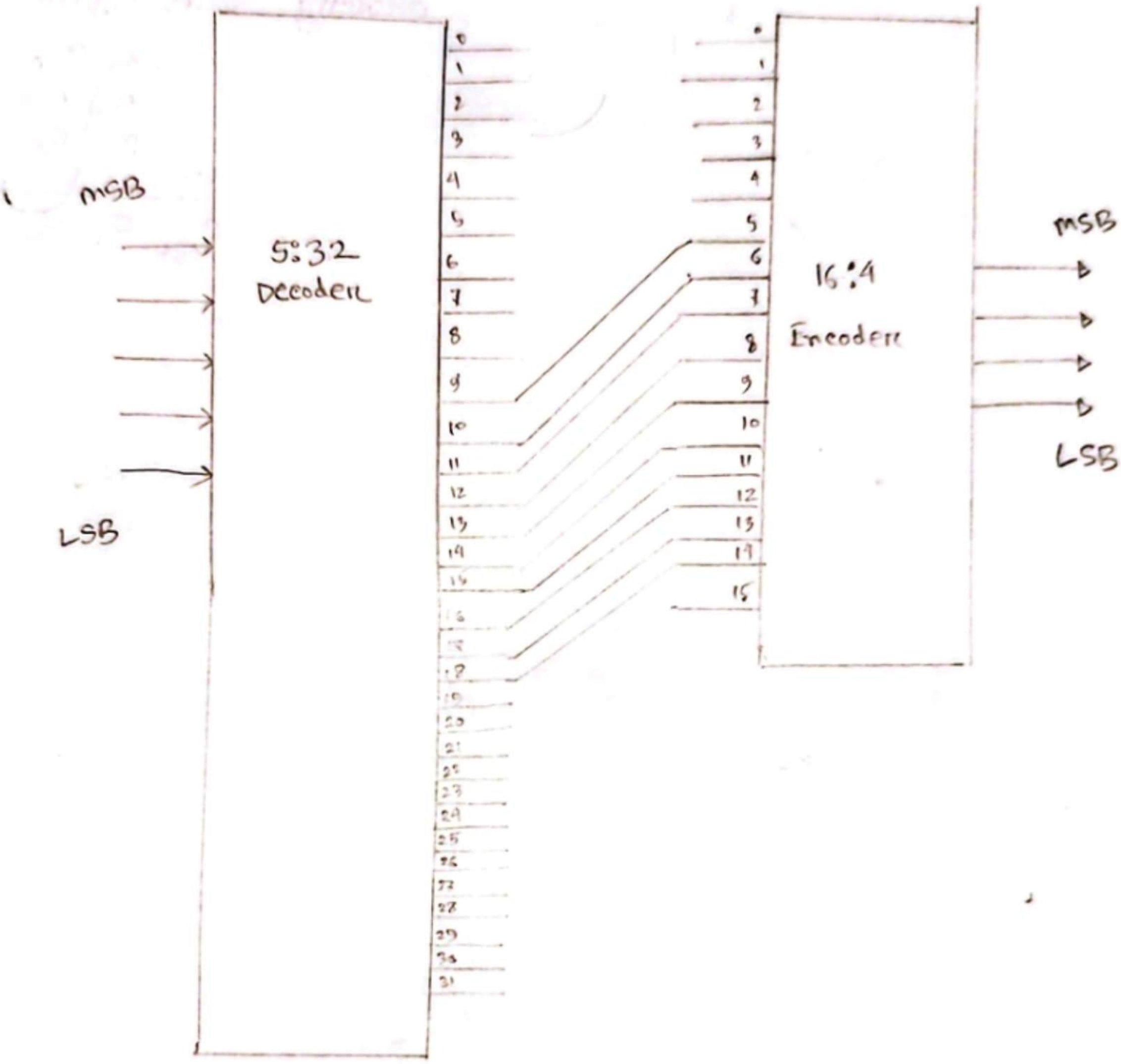


19/4/26

[Discussion]

[Answer to the Question no 1]

0 ⟶ 9 ⟶ 5



Excess-5 = Excess-9 − 4

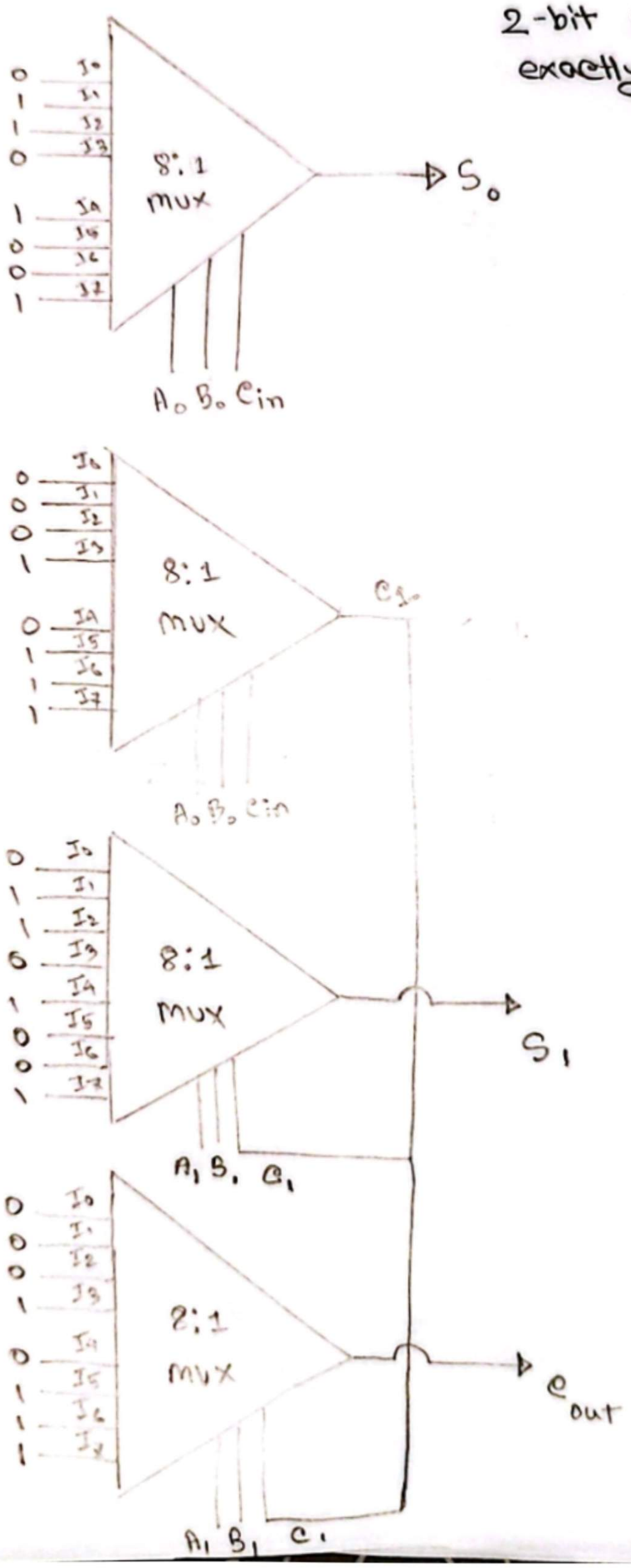
For Excess-9

⟶ Range (9 to 18)

[Ans fo the question no 2]

2-bit parallel adder using exactly four 8:1 mux(s)

	C_1	C_{in}
	A_1	A_0
(+)	B_1	B_0
C_{out}	S_1	S_0



A_0	B_0	C_{in}	C_1	S_0	
0	0	0	0	0	$\rightarrow I_0$
0	0	1	0	1	$\rightarrow I_1$
0	1	0	0	1	$\rightarrow I_2$
0	1	1	1	0	$\rightarrow I_3$
1	0	0	0	1	$\rightarrow I_4$
1	0	1	1	0	$\rightarrow I_5$
1	1	0	1	0	$\rightarrow I_6$
1	1	1	1	1	$\rightarrow I_7$

A_1	B_1	C_1	C_{out}	S_1	
0	0	0	0	0	$\rightarrow I_0$
0	0	1	0	1	$\rightarrow I_1$
0	1	0	0	1	$\rightarrow I_2$
0	1	1	1	0	$\rightarrow I_3$
1	0	0	0	1	$\rightarrow I_4$
1	0	1	1	0	$\rightarrow I_5$
1	1	0	1	0	$\rightarrow I_6$
1	1	1	1	1	$\rightarrow I_7$